

**MACHINE LEARNING MODEL DEVELOPMENT & PERFORMANCE
EVALUATION**

Customer Churn Prediction for ABC Communications Ltd

Machine Learning Internship Programme
AnalystLab Africa

Week 3 Assignment

Prepared by: Christian Agyapong

Table of Contents

Table of Contents

1. Executive Summary2

2. Evaluation Dataset2

3. Evaluation Metrics3

5. Model Performance Comparison5

6. Logistic Regression Evaluation5

7. Decision Tree Evaluation6

8. Random Forest Evaluation8

9. XGBoost Evaluation9

10. ROC Curve Analysis 10

11. Confusion Matrix Comparison 10

12 Overall Model Ranking 11

13. Model Selection 12

14. Business Interpretation 13

15. Recommendations 13

Conclusion..... 14

1. Executive Summary

Customer churn is an important business challenge for telecommunications companies because losing existing customers can negatively affect revenue, customer lifetime value, and long-term business growth.

Following the data understanding and preprocessing activities completed during Weeks 1 and 2, this project focused on developing machine learning classification models capable of predicting whether a customer is likely to churn.

The prepared Telco Customer Churn dataset was used to train and evaluate four supervised machine learning algorithms:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost

The models were evaluated using **Accuracy, Precision, Recall, F1-score, and ROC-AUC**, together with confusion matrices and ROC curve analysis.

Among the four models, **XGBoost achieved the highest overall accuracy of 80.27% and ROC-AUC of 84.46%**. Logistic Regression achieved a comparable ROC-AUC of 84.24%, while Decision Tree achieved the highest F1-score of 59.89%.

Based on the overall performance, **XGBoost is recommended as the primary candidate for deployment**, subject to additional validation and business cost analysis.

The analysis also showed that contract type, particularly **month-to-month contracts**, tenure group, internet service type, and online security were among the most influential predictors of customer churn.

2. Evaluation Dataset

The processed dataset was divided into training and testing datasets.

DATASET	ROWS	FEATURES
TRAINING	5,634	50
TESTING	1,409	50

The target variable was:

- 0 — Customer does not churn
- 1 — Customer churns

Training distribution

<i>Class</i>	<i>Customers</i>
<i>No Churn</i>	4,139
<i>Churn</i>	1,495

Testing distribution

<i>Class</i>	<i>Customers</i>
<i>No Churn</i>	1,035
<i>Churn</i>	374

The distribution demonstrates that the dataset is moderately imbalanced toward the non-churn class.

3. Evaluation Metrics

1 Accuracy

Accuracy measures the proportion of all predictions that were classified correctly.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Although useful, accuracy should not be considered independently because of the class imbalance.

3.2 Precision

Precision measures the proportion of customers predicted to churn who churned.

$$Precision = \frac{TP}{TP + FP}$$

A high precision means that the company is less likely to spend retention resources on customers who were incorrectly classified as potential churners.

3.3 Recall

Recall measures the proportion of actual churners successfully identified by the model.

$$Recall = \frac{TP}{TP + FN}$$

Recall is particularly important for this project because failing to identify a customer who subsequently churns represents a missed retention opportunity.

3.4 F1-score

F1-score provides a balance between precision and recall.

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

It is useful when both false positives and false negatives matter.

3.5 ROC-AUC

ROC-AUC measures how well a model distinguishes between churn and non-churn customers across different classification thresholds.

A value closer to 1.0 indicates stronger discriminatory ability.

5. Model Performance Comparison

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.799148	0.653199	0.518717	0.578241	0.842419
1	Decision Tree	0.798439	0.634731	0.566845	0.598870	0.829844
2	Random Forest	0.777147	0.602740	0.470588	0.528529	0.819997
3	XGBoost	0.802697	0.661074	0.526738	0.586310	0.844630

6. Logistic Regression Evaluation

Logistic Regression achieved:

Accuracy: **79.91%**

Precision: **65.32%**

Recall: **51.87%**

F1-score: **57.82%**

ROC-AUC: **84.24%**

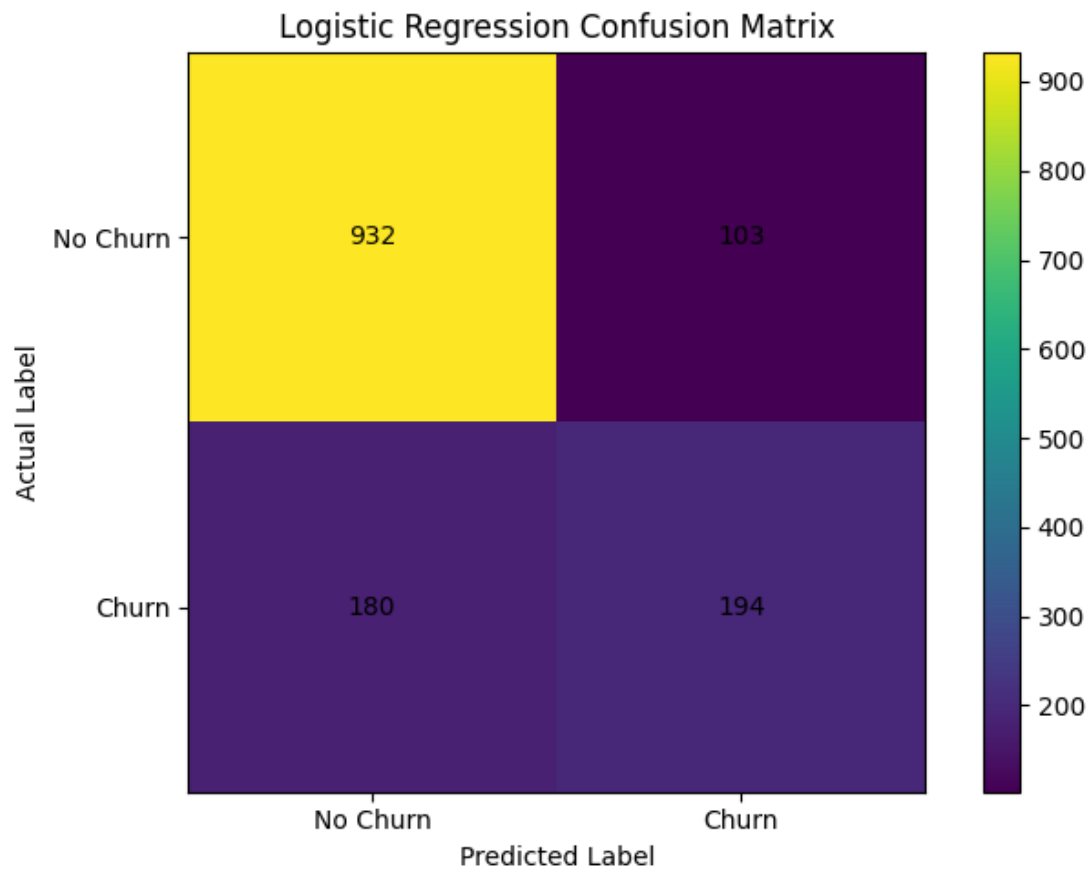
Logistic Regression Classification Report

```
=====
```

	precision	recall	f1-score	support
No Churn	0.84	0.90	0.87	1035
Churn	0.65	0.52	0.58	374
accuracy			0.80	1409
macro avg	0.75	0.71	0.72	1409
weighted avg	0.79	0.80	0.79	1409

The model performed strongly despite being relatively simple.

Its ROC-AUC of 84.24% indicates good discrimination between churn and non-churn customers.



7. Decision Tree Evaluation

Decision Tree achieved:

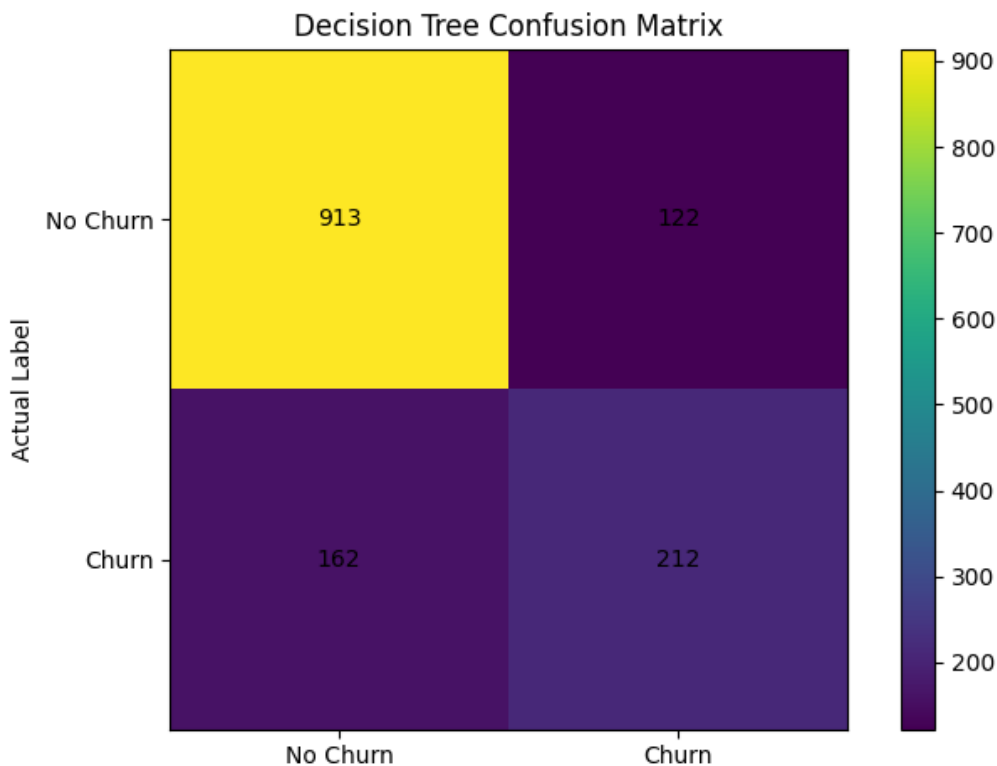
- Accuracy: **79.84%**
- Precision: **63.47%**
- Recall: **56.68%**
- F1-score: **59.89%**
- ROC-AUC: **82.98%**

The Decision Tree achieved the **highest recall and F1-score** among the four models.

Decision Tree Classification Report				
	precision	recall	f1-score	support
No Churn	0.85	0.88	0.87	1035
Churn	0.63	0.57	0.60	374
accuracy			0.80	1409
macro avg	0.74	0.72	0.73	1409
weighted avg	0.79	0.80	0.79	1409

The model correctly identified 212 actual churners, the highest number among the evaluated models.

This makes the model particularly interesting from a customer-retention perspective.



8. Random Forest Evaluation

Random Forest achieved:

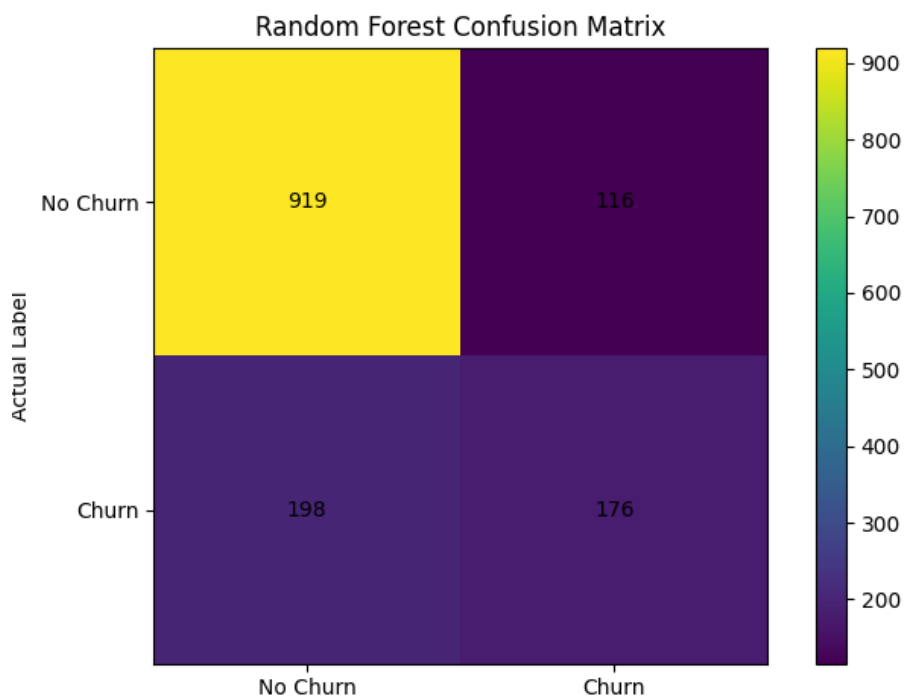
- Accuracy: **77.71%**
- Precision: **60.27%**
- Recall: **47.06%**
- F1-score: **52.85%**
- ROC-AUC: **82.00%**

Random Forest Classification Report

	precision	recall	f1-score	support
No Churn	0.82	0.89	0.85	1035
Churn	0.60	0.47	0.53	374
accuracy			0.78	1409
macro avg	0.71	0.68	0.69	1409
weighted avg	0.76	0.78	0.77	1409

.

Confusion Matrix



9. XGBoost Evaluation

XGBoost achieved:

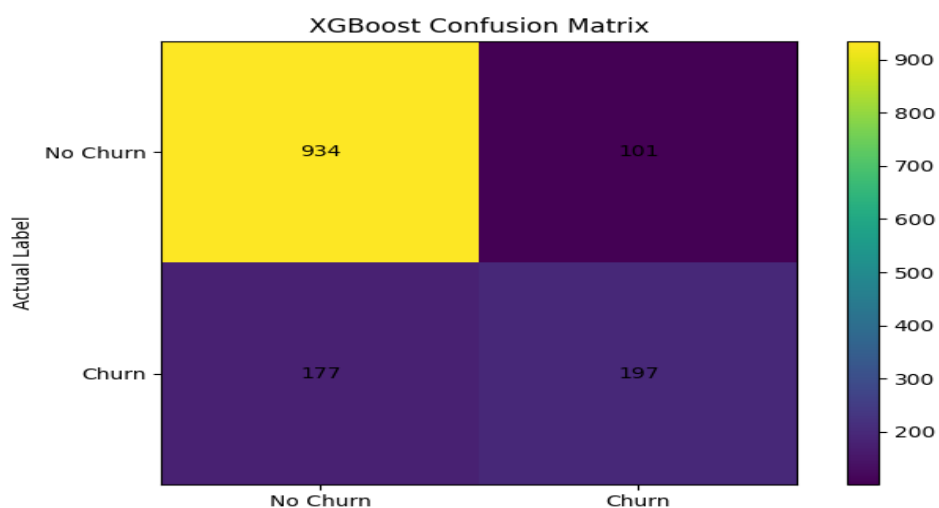
- Accuracy: **80.27%**
- Precision: **66.11%**
- Recall: **52.67%**
- F1-score: **58.63%**
- ROC-AUC: **84.46%**

XGBoost achieved the **highest accuracy, precision, and ROC-AUC.**

XGBoost Classification Report

	precision	recall	f1-score	support
No Churn	0.84	0.90	0.87	1035
Churn	0.66	0.53	0.59	374
accuracy			0.80	1409
macro avg	0.75	0.71	0.73	1409
weighted avg	0.79	0.80	0.80	1409

Confusion Matrix

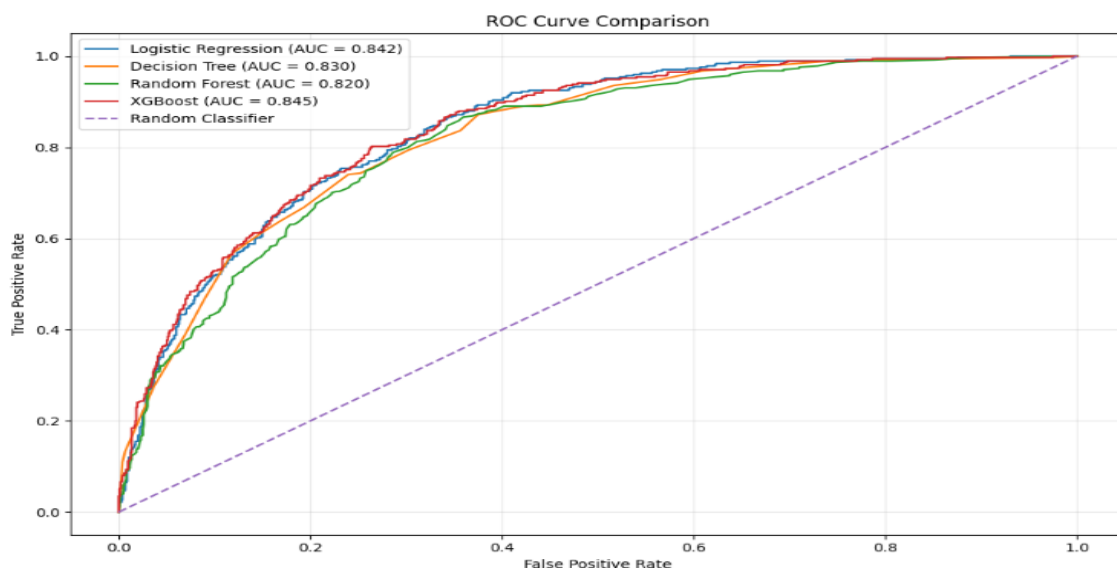


10. ROC Curve Analysis

ROC curves were generated to compare the ability of the models to distinguish between churn and non-churn customers.

The ROC-AUC scores were:

Model	ROC-AUC
XGBoost	0.8446
Logistic Regression	0.8424
Decision Tree	0.8298
Random Forest	0.8200



11. Confusion Matrix Comparison

The confusion matrices reveal an important difference between the models.

Model	True Negatives	False Positives	False Negatives	True Positives

Logistic Regression	932	103	180	194
Decision Tree	913	122	162	212
Random Forest	919	116	198	176
XGBoost	934	101	177	197

12 Overall Model Ranking

Considering the complete set of evaluation metrics:

1. XGBoost

Best overall model

- Highest accuracy
- Highest precision
- Highest ROC-AUC
- Strong F1-score
- Good recall

2. Logistic Regression

Strong baseline

- Very competitive ROC-AUC
- Good accuracy
- Easier to interpret
- Lower computational complexity

3. Decision Tree

Best recall/F1 performance

- Highest recall

- Highest F1-score
- Strong ability to identify churners
- Lower ROC-AUC than XGBoost and Logistic Regression

4. Random Forest

Weakest performance in this experiment

- Lowest accuracy
- Lowest recall except relative comparison
- Lowest F1-score
- Lowest ROC-AUC

13. Model Selection

Based on the overall evaluation, **XGBoost is selected as the preferred model.**

The main reasons are:

1. Highest accuracy — **80.27%**
2. Highest precision — **66.11%**
3. Highest ROC-AUC — **84.46%**
4. Strong F1-score — **58.63%**
5. Good recall — **52.67%**

However, the Decision Tree should not be dismissed because it achieved the highest recall of **56.68%**.

If ABC Communications Ltd. considers missing a potential churner to be substantially more costly than contacting a customer who does not churn, then the company may prefer a model or classification threshold optimized specifically for recall.

14. Business Interpretation

The evaluation demonstrates that machine learning can identify meaningful patterns associated with customer churn.

However, the models do not identify every customer who eventually churns.

For example, XGBoost produced **177 false negatives**.

These customers actually churned but were predicted as non-churners.

This represents a potential business risk because the company may fail to intervene before those customers leave.

Therefore, future deployment should consider:

- Cost of false negatives
- Cost of false positives
- Customer lifetime value
- Retention campaign cost
- Appropriate classification threshold

Model selection should ultimately be based on **business impact rather than accuracy alone**.

15. Recommendations

Based on the evaluation, the following recommendations are proposed:

Recommendation 1 — Use XGBoost as the initial deployment candidate

XGBoost demonstrated the strongest overall predictive performance.

Recommendation 2 — Optimize the classification threshold

The default 0.5 threshold should not automatically be considered optimal.

A lower threshold could potentially identify more churners and improve recall.

Recommendation 3 — Evaluate business costs

ABC Communications should quantify the cost of:

- Missing a churner
- Contacting a non-churner
- Providing retention incentives
- Losing a customer

Recommendation 4 — Monitor model performance

Once deployed, the model should be monitored regularly for:

- Accuracy changes
- Recall changes
- Data drift
- Customer behavior changes
- Model degradation

Conclusion

Four machine learning classification algorithms were successfully developed and evaluated for customer churn prediction.

The results show that **XGBoost achieved the strongest overall performance**, with an accuracy of **80.27%** and ROC-AUC of **84.46%**.

Logistic Regression also performed competitively, while Decision Tree achieved the highest recall and F1-score.

Random Forest produced the lowest overall results in this experiment.

The analysis demonstrates that no single metric is sufficient for selecting a churn prediction model. Although XGBoost is recommended based on overall performance, the business should conduct threshold optimization and cost-sensitive analysis before final deployment.

The evaluation therefore provides a strong foundation for moving the customer churn prediction project toward production-oriented machine learning.